RuyiSDK - A Unified and Extensible Platform for RISC-V Ecosystem Development

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Abstract

The RISC-V instruction set architecture is evolving rapidly, and new profiles often render existing commercial chips non-compliant. This drives vendors to revise processor core designs more frequently while making it harder to maintain software support for legacy chips. RuyiSDK is a project launched to address these challenges. By collaborating with vendors, RuyiSDK develops Linux distributions that support both new and older chips, upstreams toolchain support to reduce the burden of maintaining vendor-specific forks, and establishes a unified platform to consolidate development resources. This platform is the Ruyi Package Manager, which integrates customized systems images, toolchains, and other resources into a cohesive framework. This paper presents the design of the Ruyi Package Manager and the Ruyi Packages Index, including three key features, package management, the Ruyi Virtual Environment, and the RuyiSDK Device Provisioning Wizard. The Ruyi Package Manager has implemented most of its core functionality, providing broad support for existing RISC-V chips and lowering the entry barrier to RISC-V software development.

Introduction and Motivation

The RISC-V instruction set architecture is evolving rapidly, and the announcement of new profiles often renders already-deployed chips non-compliant. This drives vendors to continuously iterate on processor core designs and release new development boards, fostering a highly diverse ecosystem. For development boards based on specific SoCs, achieving optimal performance often requires customized toolchains, firmware, and operating systems. However, many RISC-V board manufacturers have limited capacity for long-term software support. Without sufficient motivation to upstream their work, older platforms eventually lose active maintenance and fall out of the software ecosystem.

RuyiSDK is a project launched to address these challenges, aiming to provide a comprehensive, one-stop solution tailored for RISC-V developers. It collaborates with chip vendors and board manufacturers, optimizing foundational software, promoting adaptation of unsupported applications, and cultivating a vibrant developer community. RuyiSDK maintains forks of GCC and LLVM, initially providing support for vendor-defined extensions, and progressively pushing patches upstream. It also develops Linux distributions such as RevyOS, based on Debian, to adapt to and continuously support all development boards based on Xuantie CPUs. The long-term goal is to ensure that all RISC-V development boards can rely on a stable and upstream-evolving software ecosystem, and to expose that ecosystem to developers and end users through a unified platform.

This platform is the Ruyi Package Manager, which integrates customized system images, toolchains, and other resources into a cohesive framework, incorporating contributions not only from the RuyiSDK project but also from external organizations. This paper presents the design of the Ruyi Package Manager and the Ruyi Packages Index, emphasizing three key features, namely package management, the Ruyi Virtual Environment and the RuyiSDK Device Provisioning Wizard.

Methodologies

Unlike conventional package managers such as APT and Pacman, which focus on user-level software packages, the Ruyi Package Manager also handles uncommon assets such as firmware, system images, and source code. Ruyi packages fall into six main categories: toolchain, emulator, source tarball, board image, analyzer, and extra packages. These packages may vary in compatibility or usage across different SoCs or MCUs, requiring detailed records of ISA features for each CPU core. By linking development boards to their respective CPU ISAs, the system can automatically pass architecture-specific parameters or block incompatible operations during package execution.

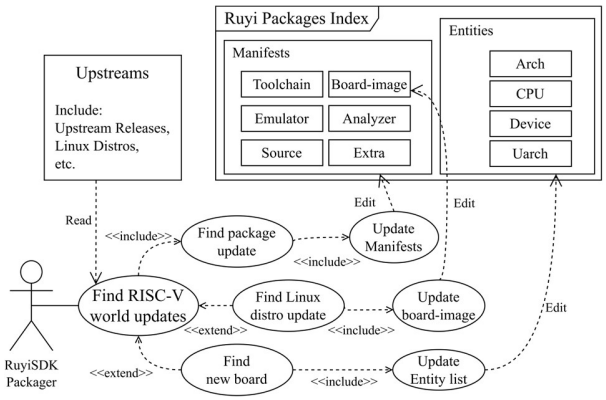

Figure 1: Design of Ruyi Packages Index.

The Ruyi Packages Index is composed of two main components， the entity list and TOML-based package definitions. Its design is shown in Figure 1. The entity list records detailed information for each board model, including available hardware variants, the CPUs or MCUs used, and their supported ISA features. The package definitions specify metadata such as name, description, download URLs, and hashes. Since upstream versioning schemes are inconsistent, the Ruyi Packages Index maps all versions to the semantic versioning standard, while preserving original upstream versions as string metadata. Maintenance of Ruyi Packages and the entity list is handled through a combination of manual contributions and automated tooling, similar to how package repositories are maintained in a Linux distribution. Notably, the Ruyi Packages Index consists entirely of text files, which makes it Git-friendly and easy to manage without redistributing vendor binaries.

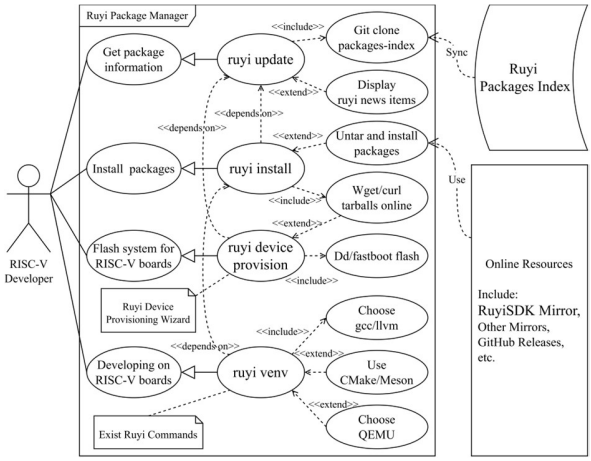

Figure 2: Design of Ruyi Package Manager.

The design of the Ruyi Package Manager is illustrated in Figure 2. The Ruyi Package Manager operates based on metadata from the Ruyi Packages Index. As a package manager, it clones the Ruyi Packages Index to the local system or updates it based on an existing cache, then select and download the required packages online and perform offline development。 The Ruyi Virtual Environment can automatically select suitable toolchains and emulators based on the specified board and build target, and automatically pass in parameters such as -mcpu, -march, or -mabi. The Device Provisioning Wizard can also list available operating systems and flash the latest firmware and system images based on the board model.

In a typical use case, when a user receives a new RISC-V development board, the Ruyi Device Provisioning Wizard provides an interactive Q&A interface that helps users flash the latest system images and firmware onto their devices and access relevant development resources. If a development board can be connected to the host machine in a supported manner and accurately identified, the wizard is able to recognize the board and list the latest available firmware and system images online. Alternatively, the user can manually specify the board model to initiate the process. Once the development board has been updated and successfully booted, developers can use the Ruyi Package Manager to install the appropriate toolchain, create a Ruyi Virtual Environment, and compile software for their board.

Result

The Ruyi Package Manager has already implemented its core functions and is still under rapid development and iteration. It already has tens of thousands of downloads, supports most Linux distributions via Nuitka binaries, and provides over 75 ruyi packages. Regarding upstream integration, some toolchains maintained by RuyiSDK provide support not yet fully available in upstream projects, for example, gnu-plct-rv64ilp32-elf supports the rv64ilp32 ABI, and llvm-plct includes support for the xtheadvector extension. From the vendor perspective, RuyiSDK has already achieved comprehensive support for RISC-V development boards from MilkV and Sipeed, offering not only dedicated toolchains, emulators, and system images, but also source code demos.

The RuyiSDK Device Provisioning Wizard provides more than 52 images for more than 31 development boards, and the number is still increasing. In recent versions of the Ruyi Virtual Environment, manual selection is still required for the toolchain and emulator used within a virtual environment, but future versions will utilize the entity list to assist with this process. The Ruyi Packages Index already includes entity lists for 33 devices and 19 micro-architectures, ranging from the generic rv64gc, SiFive U74, Xuantie C910, to less common architectures such as Xiangshan Nanhu. While already suitable for Linux software development, RuyiSDK has also begun supporting embedded development boards released by WCH.

Some operations require root privileges, but the Ruyi Package Manager will only invoke sudo after obtaining the user's consent, respecting user control and system security. Another challenge in promoting RISC-V boards is that closed-source software is often released only for x86_64 architecture. The Ruyi Package Index offers Box64 packages, which can run with systemd-binfmt, enabling users to easily configure environments for running Linux applications across architectures or run Windows applications via Wine on RISC-V systems.

To enhance customizability, package category names are configurable, and the package index URL is also configurable, allowing users to manage their own packages and repositories using Ruyi. The Ruyi Package Manager includes commands to assist with repository management and can be used on any architecture, as it is developed entirely in pure Python. Both the package manager and the package index support plugins, providing further extensibility.

Conclusion

The Ruyi Package Manager has demonstrated wide compatibility across platforms and distributions, supporting both native and cross-development for RISC-V CPUs, SoCs, and MCUs. Its powerful features, such as virtual environments, device provisioning, and user customizability, make it a practical and scalable platform for the growing RISC-V software ecosystem. The Ruyi Package Manager and its components have already accumulated tens of thousands of downloads and over a thousand users. As features continue to improve, the user base is expected to keep growing. It enables developers to focus on innovation and helps expand the adoption of RISC-V, contributing to its development as a mainstream architecture.